

A new probabilistic coverage model for ambulances deployment with hypercube queuing approach

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Abstract Aiming at improving the efficiency and reliability of ambulance service, several location models for ambulance stations have been proposed in the facility location literature. Two well-known approaches to this problem are coverage and median models. Coverage model looks for the location to maximize the (deterministic or probabilistic) covered demand of ambulance calls. Hence, this model can be thought of reliability-oriented model. In median model, the objective is to minimize the total traveling distance of the ambulances from the station to the scene of call and thus is the cost-oriented model. In this paper, a new probabilistic coverage model is introduced. It uses the idea of MEXCLP and mixes it with the hypercube queuing model. The model is applied in a real case for one of Tehran city's zones in Iran. The results were satisfactory and show the applicability of the model. Sensitivity analysis was conducted to determine the overall behavior of system responsiveness against parameters of the model. Discussions on sensitivity analysis are presented.

Keywords Emergency service vehicles · MEXCLP · Hypercube queuing model · Ambulance deployment

1 Introduction

A major concern of emergency medical service centers is to provide first care medical assistance to the emergency calls as soon as possible in a best way. The time elapsed between an emergency call and its service, called response time, along with the number of ambulance dispatched and fulfilled the service are the two important performance measures in these types of problems. Since it is often difficult and expensive to

experiment with an actual system, mistakes are costly both in money and in potential mortality. Collecting data to verify a good system might take months of data gathering, hence instead of experimenting on the actual system, operations research professionals generally build models of systems that can be implemented virtually on a computer. System errors can be found on the model before they are implemented on the actual system. It is generally well worth the cost of building the model, collecting data, and running the model as opposed to trying to experiment on the actual system [1]. Several studies are found in the literature suggesting approaches to rationalize the importance of optimal usage of available capacities and improve client service. By the way, a number of papers do not actually consider the probabilistic nature of request calls and service processes, and this fact that ambulances surely are not always available for servicing a call [2]. As mentioned by Takeda et al. [2], examples of probabilistic approaches for ambulance deployment appear in Brandeau and Larson [3], Eaton et al. [4], Goldberg et al. [1], and Fujiwara [5].

The hypercube queuing model developed by Larson which extended by other researchers is an effective analytical model for planning server-to-client systems. Given a system arrangement, it is able to evaluate a variety of performance measures appropriate for decision making. In fact, it is not an optimization model which determines the optimal configuration for the system, but it can provide a reasonably complete evaluation for each proposed arrangement. It can also be incorporated in optimization approaches to deal with probabilistic location problems. For instance, Batta et al. [6] suggested its use in an iterative procedure to relax the assumption of independence in the maximum expected covering location problem, MEXCLP (Daskin [7]). Chiyoshi et al. [8] analyzed nonhomogeneous servers and compared MEXCLP and the adjusted maximum expected covering location problem, AMEXCLP. Saydam and Aytug [9] proposed a genetic algorithm that integrates MEXCLP with a hypercube approximation algorithm developed by Jarvis [10] in order to solve MEXCLP with increased

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accuracy. Gendreau et al. [11] used tabu search in similar outline. Galvão et al. [12] applied simulated annealing in the solution of MEXCLP and the maximum availability location problem, MALP [2, 13]. As mentioned by Goldberg [1], Church and Weaver [14] extend the maximal coverage model and represent the “vector assignment p -median model.” Hogan and ReVelle [15] extend the maximal covering model and the set covering model by adding a second objective to maximize the number calls covered by two or more vehicles. They combine the objectives using weighting factors and evaluate how the solution alters as the weight of each two objectives changes. Gendreau, Laporte, and Semet [16] developed a search for a model that maximizes the number of double-covered calls. Pirkul and Schilling [17] model the objective of maximizing covered calls subject to restrictions on the number of received calls that each vehicle can response (as measured by the number of calls, it may serve in the study horizon) and a requirement for backup coverage. Narasimhan, Pirkul, and Schilling [18] expand the model to include the objective of maximizing the number of responded calls of different service levels. They include constraints on the number of vehicles, the call capacity of each vehicle, and determining the coverage for different levels of service. Marianov and ReVelle [19] extend the set covering model to ensure that each demand has a vehicle actually available within the standard time, with a specific probability p . The objective of minimizing the required number of servers remains the same. Queuing theory is applied to the development of the availability constraints so that one does not need to assume that vehicles are independent as was done in ReVelle and Hogan [20] and Daskin. Instead, the model uses the zone-based busy assumption. Silva and Serra [21] extend the Marianov and ReVelle work to include multiple call priorities. Marianov and ReVelle [22] extend ReVelle and Hogan [20] by using queuing theory to develop the required coverage constraints. The zone-based busy assumption is still required. Marianov and Serra [23] extend the work of Marianov and ReVelle to include decisions on base location and on allocating customers to centers. The bases behave as queuing systems, so calls may wait in line. Marianov and Serra [24] use the techniques in their work and extend the set covering model to include minimizing the number of facilities and the number of vehicles. Constraints on queue length are also considered. In all of the Marianov papers in this paragraph, all demands assigned to an EMS center must assign to that center, regardless of the how busy that center is. This restricts the validity of these models in an emergency situation.

2 Literature review and motivation for the research

“Hypercube model,” first introduced by Larson [30] for evaluating the performance of a set of base locations. The approximate hypercube (A-hypercube) models first introduced by

Larson [31] and later extended by Jarvis [26] have more realistic assumptions about the behavior of the system than the original hypercube model. Benveniste [25] extended A-hypercube by adding the objective of minimizing system travel distance and developing in an optimization strategy that selects server locations and service areas (what zones are first choice for each zone). Jarvis [26] extended A-hypercube by including general call service times. This enables the modeling of systems where the call service times depend both on the serving vehicle and the demand zone. Brandeau and Larson [3] described applying the hypercube model to locate ambulances in Boston. Batta, Dolan, and Krishnamurthy [6] combined the ideas in A-hypercube with those in Daskin's expected covering model. This new model remedies the independence assumption in Daskin's model by including Larson's Q-factors in the objective and the authors build a heuristic to find good sets of locations. Goldberg and Paz [27] extended Jarvis' model by adding the objective of maximizing the expected number of calls covered (they estimate the probability that travel is less than the coverage time standard) and by embedding the new model in a location selection heuristic. The assumption of a fixed vehicle preference ordering for each zone remains. Morabito and Mendonca [28] use the hypercube model to consider the deployment of ambulances in Brazil. Geroliminis et al. [34] developed a model for locating emergency vehicles on urban networks based on a generalization of the hypercube queuing model. They considered spatial and temporal characteristics of demand such as the possibility that a server is not always available when its service is required, and the service rates are not identical and may be variable both among servers and incident characteristics. The model was applied for locating tow-away vehicles on the I-80 freeway in the San Francisco Bay area.

Recently, Takeda et al. [2] studied the application of the hypercube queuing model to SAMU-192, the urban Emergency Medical Service of Campinas in Brazil. They analyzed different decentralization scenarios, and the impacts of ambulance decentralization were evaluated in terms of mean travel times. No mathematical model was formulated for ambulance deployment. Budge et al. [29] showed that the AH model outperforms exact hypercube model and simulation approaches, and in terms of computational time are relatively insensitive to system characteristics, and they are sufficiently accurate for many practical purposes. Recently, Iannoni et al. [32] and Iannoni and Morabito [33] embedded hypercube model in a genetic algorithm framework to locate emergency vehicles along a highway. They extended the problem to enable multiple dispatches (e.g., more than one server can intervene for the same incident). By embedding the hypercube into a genetic algorithm, they supported decisions, such as determining the optimal districts for the system in order to optimize the mean performance measures. Geroliminis et al. [35, 36] integrated the location and distracting decisions in the same optimization and solve the problem by

using steepest descent (Geroliminis et al. [35]) and genetic algorithms (Geroliminis et al. [36]).

Table 1 summarizes the literature review and demonstrates special features of this study. Only works are reviewed that are most related to this study, i.e., those that used hypercube queuing model in ambulance location and deployment. As this summary shows, this study is the only one that considers ambulance deployment using hypercube queuing model and simultaneously optimizes expected covering. Other similar works either decide on ambulance location not deployment or were just descriptive models and did not offer any optimization model for deployment.

3 Problem definition

In this paper, as mentioned before, we tried to mix the ideas of Daskin and Larson and introduce a new model. If we want to state simply, we used the availability probability of queuing model and the average requests arrival of Poisson process in MEXCLP model, then the solution will be our service planning in EMS center. Doing this, we considered two types of EMS vehicles, similar to the work done in [2]; for basic needs,

we had ambulances called basic service vehicle (BSVs), and for advanced needs like serious and critical scorches or some kind of heart attacks, we had advance service vehicles (ASVs). The area should be covered with EMS center, divided to some regions. Then we categorized the types of needs each region could be facing. Type one need is a request for just one BSV, type two is the request for more than one BSV, and so on for ASVs; after that, we had the mixed type request in which a BSV and an ASV is requested simultaneously and the last request type was the former one but more than one at least from one type. After request types categorization, we labeled a number to each ambulance called preference number, which means that dispatching ambulances to regions would be different. For example, the BSV 1 preference to region 2 might be 1 and to region 1 might be the second. This preference can be a real-world assumption that has been used in many models like [2]. Probably, we can attribute this preference number to the driver familiarity with that region. Then we started to create the steady-state equations with hypercube model assumptions.

Independent poisson arrivals Emergency calls of each region for every type were generated as a Poisson process, independently from the others.

Table 1 Literature review summary

Reference	Models and features	Decisions	Solution approach
Benveniste [2]	A-hypercube, minimizing system travel distance	Ambulance location, service areas	NLP, augmented Lagrangian method
Jarvis [10]	A-hypercube, vehicle and zone depended service times	Approximating performance of the system	No optimization model
Brandeau and Larson [3]	Hypercube model	Ambulance location	Locate-allocate heuristics
Batta, Dolan, and Krishnamurthy [6]	A-hypercube in Daskin's MEXCLP	Ambulance location	Heuristics
Goldberg and Paz [27]	A-hypercube, maximum expected covering	Ambulance location	Heuristics
Morabito and Mendonca [28]	Hypercube model	Ambulance deployment on highways	No optimization model/scenario planning
Geroliminis et al. [34]	Generalized hypercube model, minimizing mean response time	Ambulance location on networks	Random search followed by the steepest descent method
Takeda et al. [2]	Hypercube model, mean travel times criteria	Ambulance decentralization	No optimization model/no solution approach
Iannoni and Morabito [33]	Multiple dispatch and partial backup hypercube model	Ambulance location on highways	No optimization model/no solution approach
Iannoni et al. [32]	Hypercube model, optimize the mean performance measures	Optimal districts	Genetic algorithm coded in Pascal + a simple enumerative algorithm (exact)
Geroliminis et al., [35]	Hypercube model	Ambulance location and districting	steepest descent algorithm
Budge et al. [29]	A-hypercube, probability-of-coverage map	Travel-time distribution model	No optimization model/no solution approach
Geroliminis et al., [36]	Hybrid hypercube–genetic algorithm	Ambulance location and districting	Genetic algorithm
This paper	Hypercube model + Ideas form Daskin's MEXCLP, maximum expected covering	Ambulance deployment	Exact (ILP solvers in Lingo)

Server locations All ambulances, when available, remained in the center waiting for request call.

Fixed-preference dispatching There is a list of ambulance dispatching preferences for each region.

Service times Service times were exponentially distributed with means μ and μ' for BSVs and ASVs separately.

If we ignore the initial installation of center and consider the fact that the system operates in steady-state situation, each possible state of the system is represented by a vector with N component (ambulances) assuming values 0 or 1 (0 for free and 1 for busy). Therefore, there will be 2^N possible states in the system. Finding the solution for equations, we replace the solution in the objective function. As it is obvious, using hypercube queuing model is solely just for finding every state probability and other aspects of hypercube model, such as evaluating the system configuration performance is ignored. If we want to clarify the situation laconically, our approach is to find the probability of each state with hypercube model, then use it in the MEXCLP model where the availability probability of each ambulance to dispatch is needed. After creating the objective function and the constraints, we can solve the model and reach the exact solution using OR problem solvers softwares like LINGO. In the next section, we will explain how we aggregated the idea of the two known EMS models and made our desired analytical model, and then we will have some solution analysis and a conclusion at last.

4 Mathematical model

Deployment planning in the maximum covering problems is formulated as an integer linear programming (ILP) problem including these features as follows:

Indices:

$a, b, \dots, u$	Number of the BSVs
$v, w, \dots, z$	Number of the ASVs
i	Number of the dispatched BSV $\in \{a, b, \dots, u\}$
k	Number of the dispatched ASV $\in \{v, w, \dots, z\}$

Decision variables

Y_{ij}	Binary variable related to the covered region with a BSV; if region j would be covered with BSV i , then Y_{ij} is equal to one, otherwise it is zero
$Y_{a, \dots, u, j}$	Binary variable related to the covered region with k BSVs ($k > 1$); if the region j would be covered with k BSVs, then $Y_{a, b, c, \dots, u, j}$ is equal

Y'_{kj}	Binary variable related to the covered region with an ASV; if region j would be covered with ASV i , then Y'_{kj} is equal to one, otherwise it is zero
$Y'_{v, \dots, z, j}$	Binary variable related to the covered region with k ASVs ($k > 1$); if the region j would be covered with k ASVs, then $Y'_{v, \dots, z, j}$ is equal to one, and the k numbers of the dispatched indices would be equal to one
$Y''_{a, b, j}$	Binary variable related to the covered region with a BSV and an ASV; if region j would be covered with the BSV a and ASV b , then $Y''_{a, b, j}$ is equal to one, otherwise it is zero
$Y''_{a, \dots, u, v, \dots, z, j}$	Binary variable related to the covered region with k BSVs and k' ASVs ($k \times k' > 1$); if the region j would be covered with k BSVs and k' ASVs then $Y''_{a, \dots, u, v, \dots, z, j}$ is equal to one, and the k numbers of the dispatched indices from a to u would be equal to one and k' numbers of the dispatched indices from b to w is equal to one too.

Demand parameters

λ_j	Region j demand rate for a BSV
λ_{kj}	Region j demand rate for k BSVs
λ_j	Region j demand rate for an ASV
λ_{kj}	Region j demand rate for k ASVs
λ_j	Region j demand rate for a BSV and an ASV simultaneously
$\lambda''_{kk'j}$	region j demand rate for k BSV and k' ASV simultaneously

Service rate parameters

μ	A BSV service rate
μ'	An ASV service rate

Other parameters

u	Total number of BSVs
z	Total number of ASVs
L	Number of regions

4.1 State probability parameters

The probability of each state is displayed with the p symbol in objective function, i.e., p_{0111} , which indicates the probability that we have four vehicles overall, two are BSVs and the two

others are ASVs, in which BSV 1 is free and BSV 2 is busy, and both ASVs are busy. The yellow figures indicate BSVs, the greens indicate ASVs, and the reds are the dispatched ambulances to the request regions. This differentiation in colors is used just for easiness in discrimination and has no other function.

4.2 Objective function

Max $z =$

$$\begin{aligned} & \sum_j \lambda_j \sum_{a=0}^1 \sum_{b=0}^1 \sum_{c=0}^1 \cdots \sum_{d=0}^1 P_{(0) \dots (s)(t)(u)} Y_{1,j} + \\ & \sum_j \lambda_j \sum_{a=0}^1 \sum_{b=0}^1 \sum_{c=0}^1 \cdots \sum_{d=0}^1 P_{(a)(0) \dots (s)(t)(u)} Y_{2,j} + \\ & \cdots + \\ & \sum_j \lambda_{2j} \sum_{a=0}^1 \sum_{b=0}^1 \sum_{c=0}^1 \cdots \sum_{d=0}^1 P_{(0)(b)(0) \dots (s)(t)(u)} Y_{1,2,j} + \\ & \sum_j \lambda_{2j} \sum_{a=0}^1 \sum_{b=0}^1 \sum_{c=0}^1 \cdots \sum_{d=0}^1 P_{(0)(b)(0) \dots (s)(t)(u)} Y_{1,3,j} + \\ & \sum_j \lambda_{2j} \sum_{a=0}^1 \sum_{b=0}^1 \sum_{c=0}^1 \cdots \sum_{d=0}^1 P_{(0)(b)(c)(0) \dots (s)(t)(u)} Y_{1,4,j} \\ & + \cdots + \\ & \sum_j \lambda_{2j} \sum_{a=0}^1 \sum_{b=0}^1 \sum_{c=0}^1 \cdots \sum_{d=0}^1 P_{(0)(b)(0) \dots (s)(0)(0)} Y_{u-1,u,j} \\ & \sum_j \lambda_{3j} \sum_{a=0}^1 \sum_{b=0}^1 \sum_{c=0}^1 \cdots \sum_{d=0}^1 P_{(0)(0)(0) \dots (s)(t)(u)} Y_{1,2,3,j} \\ & \sum_j \lambda_{3j} \sum_{a=0}^1 \sum_{b=0}^1 \sum_{c=0}^1 \cdots \sum_{d=0}^1 P_{(0)(0)(c)(0) \dots (s)(t)(u)} Y_{1,2,4,j} \\ & + \cdots + \\ & \sum_j \lambda'_{1j} \sum_{a=0}^1 \sum_{b=0}^1 \sum_{c=0}^1 \cdots \sum_{d=0}^1 P_{(a)(0) \dots (0) \dots (z)} Y'_{u+1,j} + \cdots \\ & \sum_j \lambda'_{2j} \sum_{a=0}^1 \sum_{b=0}^1 \sum_{c=0}^1 \cdots \sum_{d=0}^1 P_{(0)(0) \dots (0)(0) \dots (z)} Y'_{u+1,u+2,j} \\ & + \cdots + \\ & \sum_j \lambda'_{3j} \sum_{a=0}^1 \sum_{b=0}^1 \sum_{c=0}^1 \cdots \sum_{d=0}^1 P_{(a) \dots (0)(0)(0) \dots (z)} Y'_{j,1,2,3} \\ & + \sum_j \lambda'_{3j} \sum_{a=0}^1 \sum_{b=0}^1 \sum_{c=0}^1 \cdots \sum_{d=0}^1 P_{(a)(0) \dots (0)(0)(0) \dots (z)} Y'_{1,2,4,j} \\ & + \cdots + \\ & \sum_j \lambda''_{1j} \sum_{a=0}^1 \sum_{b=0}^1 \sum_{c=0}^1 \cdots \sum_{d=0}^1 P_{(0)(b) \dots (0)(y) \dots (z)} Y''_{1,1,j} + \\ & \sum_j \lambda''_{1j} \sum_{a=0}^1 \sum_{b=0}^1 \sum_{c=0}^1 \cdots \sum_{d=0}^1 P_{(0)(d) \dots (x)(0) \dots (z)} Y''_{1,2,j} \\ & + \cdots + \\ & \sum_j \lambda''_{1j} \sum_{a=0}^1 \sum_{b=0}^1 \sum_{c=0}^1 \cdots \sum_{d=0}^1 P_{(d) \dots (0)(a)(b) \dots (0)} Y''_{u,u+z,j} + \\ & \sum_j \lambda''_{2,1,j} \sum_{a=0}^1 \sum_{b=0}^1 \sum_{c=0}^1 \cdots \sum_{d=0}^1 P_{(0)(0) \dots (0)(0)(u+2) \dots (z)} Y''_{1,2,u+1,j} \\ & + \cdots + \\ & \sum_j \lambda''_{u,w,j} \sum_{a=0}^1 \sum_{b=0}^1 \sum_{c=0}^1 \cdots \sum_{d=0}^1 P_{(0)(0) \dots (0)(0) \dots (0)} Y''_{1,\dots,u,1,2,\dots,z,j} \end{aligned}$$

The objective function tries to maximize the region emergency coverage considering two factors, the availability and preference of the chosen ambulance for that region. Each term represents a portion of this function which, respectively, explains as follows:

1. First term shows covering region j with BSV number 1
2. Covering region j with BSV number 2 and so on
3. Covering the two BSV-requested region j with BSVs numbers 1 and 2

4. Covering the two BSV-requested region j with BSVs numbers 1 and 3
5. Covering the two BSV-requested region j with BSVs numbers 1 and 4 so on to...
6. Covering the two BSV-requested region j with BSVs number $u-1$ and u
7. Covering the three BSV-requested region j with BSVs numbers 1, 2, and 3
8. Covering the three BSV-requested region j with BSVs numbers 1, 2, and 4 so on to all triple combinations of BSVs
9. Covering region j with ASVs number 1 (as shown $u+1$) and so on
10. Covering the two ASV-requested region j with ASVs numbers 1 and 2 (as shown with $u+1$ and $u+2$)
11. Covering the three ASV-requested region j with BSVs numbers 1, 2, and 3
12. Covering the three ASV-requested region j with ASVs numbers 1, 2, and 4 and so on to all triple combinations of ASVs
13. Covering the two kinds of requested region j (one from each region) with BSV number 1 and ASV number 1
14. Covering the two kinds of requested region j (one from each region) with BSV number 1 and ASV number 2 and so on to all combinations of BSVs with ASVs
15. The last combination of BSVs and ASVs is shown in this term (as shown u , $u+z$)
16. Covering the two kinds of requested region j (more than one from at least one kind) with BSVs numbers 1 and 2 and ASV number 2 and so on to all combinations of two BSVs with one ASV for this request (two BSVs and one ASV)
17. Covering the two kinds of requested region j (u BSVs and w ASVs) with all we own!

4.3 Constraints

$$\sum_{i=1}^u Y_{ij} = 1 \quad \forall j \quad (1)$$

$$\sum_{k=i} Y_{a,b,\dots,j} = 1, \quad \forall j, \quad (k \text{ is the number of dispatched vehicles}) \quad (2)$$

$$\sum_{i=1}^w Y'_{kj} = 1 \quad \forall j \quad (3)$$

$$\sum_{k=i} Y'_{s,t,\dots,j} = 1, \quad \forall j, \quad (k \text{ is the number of dispatched vehicles}) \quad (4)$$

$$\sum_{i=1}^u Y''_{i,k,j} = 1 \quad \forall j \quad (5)$$

$$\sum_{k,k'=i} Y_{a,b,\dots,s,t,\dots,z,j} = 1 \quad \forall j \quad (k, k' \text{ are the same as above}) \quad (6)$$

$$\sum_{j=1}^L (Y_{ij} + Y_{i,i+1,j} + Y_{i,i+2,j} + Y_{i,i+4,j} + \dots + Y_{i,i+1,i+2,j} + \dots + Y_{i,u-1,u} + Y_{i,i+1,i+2,\dots,u,j} + Y''_{i,k,j} + \dots) \leq 1 \quad \forall i \quad (7)$$

$$\sum_{j=1}^L (Y'_{kj} + Y'_{k,k+1,j} + Y'_{k,k+2,j} + Y'_{k,k+3,j} + \dots + Y'_{k,k+1,k+2,j} + Y'_{k,w-1,w,j} + Y'_{k,k+1,k+2,\dots,w,j} + Y''_{i,\dots,u,k,\dots,w,j}) \leq 1 \quad \forall k \quad (8)$$

Constraint (1) expresses that region j must be covered just with one of the BSVs, and this BSV is chosen in order to maximize the term according to the objective function.

Constraint (2) shows that the k number requested region j must be covered just with one combination of k BSVs, and this combination is chosen in order to maximize the term according to the objective function.

Constraint (3) expresses that region j must be covered just with one of the ASVs, and this ASV is chosen in order to maximize the term according to the objective function.

Constraint (4) shows that the k number requested region j must be covered just with one combination of k ASVs, and this combination is chosen in order to maximize the term according to the objective function.

Constraint (5) expresses that the two kinds of requested region j must be covered just with one combination of a BSV and an ASV, and this combination is chosen in order to maximize the term according to the objective function.

Constraint (6) shows that the two kinds of requested region j must be covered just with one combination of k BSVs and k' ASVs, and this combination is chosen in order to maximize the term according to the objective function.

Constraint (7) assures that every BSV can just be dispatched to one region at a specified time, either dispatched alone or along with others in a combinational state.

Constraint (8) assures that every ASV can just be dispatched to one region at a specified time, either dispatched alone or along with others in a combinational state.

And the last constraint is that all of the variables are binary as discussed before.

$$\begin{aligned} Y_{ij} & \in \{0, 1\} \\ Y_{a,\dots,u,j} & \in \{0, 1\} \\ Y'_{kj} & \in \{0, 1\} \\ Y'_{v,\dots,z,j} & \in \{0, 1\} \\ Y''_{a,b,j} & \in \{0, 1\} \\ Y''_{a,\dots,u,v,\dots,z,j} & \in \{0, 1\} \end{aligned}$$

As can be seen, the resulting model is an ILP model and can be solved efficiently by standard solvers such as LINGO, LINDO, GAMS, and CPLEX. The main challenge of the

solution procedure is the calculation of state probability parameters.

4.4 Steady-state equations

Steady-state equations are one of the major parts of the planning problem. To calculate the probability of the busy or free conditions of every ambulance in each state, considering the preference order, the steady-state equations are solved, and the probability quantities gained from solved model are substituted with the p symbols defined as probability of each state in the objective function. The idea of calculating the steady-state probability equations as explained before has been obtained from Larson's hypercube model. If we have N ambulances, we will have 2^N states which represent the system state (free and busy states are two, so we will have 2^N states). But there is a difference between our model and the popular one and that is the possibility of combined requests. To handle this difference, we supposed that the overall service time when both types are dispatched simultaneously would be μ . To explain this logical assumption that it is obvious to remark that the advanced service time is shorter, so we considered μ for both. For multi-ambulance requests in each type, we considered μ , μ' for each type as we sent the vehicles simultaneously.

5 Case study

The presented model was applied to design the EMS center of one of Tehran city's zones (Tehran is a very big city that consists of almost 20 zones, in which we have selected zone 1 in the north with two BSVs and two ASVs for our model's validation). We divided this zone into four different, non-overlapping regions (North, South, East, and West). Then we started to gather the required data. We tried to collect our data randomly with this consideration that these days are not in weekends or general vacations (which means that we selected the days from working normal days), so 20 days are selected and the demand and the service parameters are measured. The sample consists of 362 requests with different types (as discussed before). Kolmogorov–Smirnov and χ^2 goodness-

ASV	1	2
region		
1	2	1
2	1	2
3	1	2
4	2	1

BSV	1	2
region		
1	1	2
2	2	1
3	2	1
4	1	2

Fig. 1 Preference matrix for BSVs and ASVs

of-fit tests were applied to verify the hypothesis of Poisson arrival, and they could not reject the hypothesis with 5 % of significance. The demand parameters will be as follows:

$$\begin{aligned}
 \lambda_1 &= 0.9 \quad (\text{calls/day}) \\
 \lambda_2 &= 0.92 \quad (\text{calls/day}) \\
 \lambda_3 &= 0.96 \quad (\text{calls/day}) \\
 \lambda_4 &= 0.94 \quad (\text{calls/day}) \\
 \lambda'_1 &= 0.7 \quad (\text{calls/day}) \\
 \lambda'_2 &= 0.65 \quad (\text{calls/day}) \\
 \lambda'_3 &= 0.82 \quad (\text{calls/day}) \\
 \lambda'_4 &= 0.72 \quad (\text{calls/day}) \\
 \lambda''_1 &= 0.3 \quad (\text{calls/day}) \\
 \lambda''_2 &= 0.4 \quad (\text{calls/day}) \\
 \lambda''_3 &= 0.25 \quad (\text{calls/day}) \\
 \lambda''_4 &= 0.3 \quad (\text{calls/day}) \\
 \lambda_{2,1} &= 0.4 \quad (\text{calls/day}) \\
 \lambda_{2,2} &= 0.42 \quad (\text{calls/day}) \\
 \lambda_{2,3} &= 0.41 \quad (\text{calls/day}) \\
 \lambda_{2,4} &= 0.35 \quad (\text{calls/day}) \\
 \lambda'_{2,1} &= 0.3 \quad (\text{calls/day}) \\
 \lambda'_{2,2} &= 0.28 \quad (\text{calls/day}) \\
 \lambda'_{2,3} &= 0.25 \quad (\text{calls/day}) \\
 \lambda'_{2,4} &= 0.24 \quad (\text{calls/day}) \\
 \lambda''_{m,n,1} &= 0.1 \quad (\text{calls/day}) \quad \forall m, n
 \end{aligned}$$

5.1 Service parameters

For the service time, the same analysis was applied and the tests could not reject the hypothesis of exponentially distributed service times with 5 % of significance. Also, variance analysis was applied to test the equality of the service times. The test could not reject the hypothesis with 10 % of significance.

$$\begin{aligned}
 \text{BSV service time } \mu &= \frac{60\text{min}}{24 \times 60} = \frac{1}{24} \text{ day} \\
 \text{ASV service time } \mu' &= \frac{48\text{min}}{24 \times 60} = \frac{1}{30} \text{ day}
 \end{aligned}$$

5.2 Preference matrix

The matrix in Fig. 1 illustrates the dispatching preference of every BSV and ASV to each region (1) in the tables which refers to first preference and (2) the second preference for each one.

Defining the steady-state equation will be our next step. In this step, we will have 16 equilibrium probability equations ($2^4=16$). Substituting any of the equations by the normalization equation (sum of all state probabilities should be equal to 1), the result is a determined system with 16 as we declared

before linearly independent equations. For instance, the equilibrium probability equation for the intermediate state 0000 (Fig. 2), where all of the ambulances are free, is as follows:

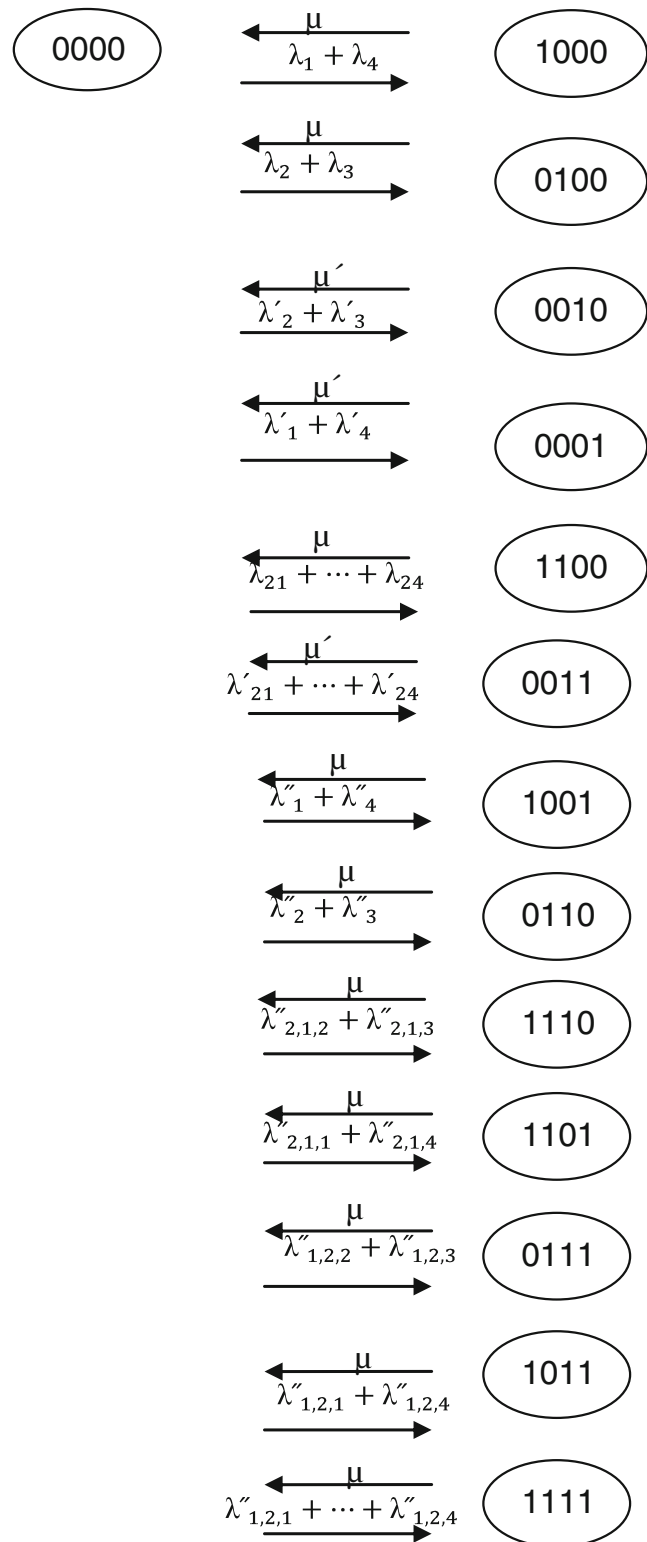


Fig. 2 Probability equation for the intermediate state 0000

Now, here is the equation in which P_{0000} is in the one side and the other states probabilities are in the other side.

$$\begin{aligned} & \left[\left(\lambda''_{2,1,1} + \lambda''_{2,1,2} + \lambda''_{2,1,3} + \lambda''_{2,1,4} \right) \right. \\ & + \left(\lambda_1 + \lambda_2 + \lambda_3 + \lambda_4 \right) + \left(\lambda_2 + \lambda_3 \right) + \\ & \left(\lambda_{21} + \dots + \lambda_{24} \right) + \left(\lambda'_2 + \lambda'_3 \right) + \\ & \left(\lambda'_1 + \lambda'_4 \right) + \left(\lambda'_{21} + \dots + \lambda'_{24} \right) + \\ & \left(\lambda''_{2,1,2} + \lambda''_{2,1,3} \right) + \left(\lambda''_{2,1,1} + \lambda''_{2,1,4} \right) + \\ & \left(\lambda''_{1,2,1} + \lambda''_{1,2,4} \right) + \left(\lambda''_{1,2,2} + \lambda''_{1,2,3} \right) + \\ & \left. \left(\lambda''_1 + \lambda''_4 \right) + \left(\lambda''_1 + \lambda''_4 \right) \right] \times P_{0000} = \\ & \left[1/\mu (P_{1000} + P_{0100} + P_{0010} + P_{0001} + P_{1100} + \right. \\ & P_{1001} + P_{0110} + P_{1110} + P_{1101} + P_{0111} + P_{1111}) + \\ & \left. 1/\mu (P_{0011} + P_{0010} + P_{0001}) \right] \end{aligned}$$

Other equations will be like the former one. We used MATLAB R2008b software to solve the steady-state equations. The solution is as follows:

$$\begin{aligned} P_{0000} &= 0.0273 \\ P_{1000} &= 0.0219 \\ P_{0100} &= 0.0288 \\ P_{0010} &= 0.0294 \\ P_{0001} &= 0.0268 \\ P_{1100} &= 0.0473 \\ P_{1010} &= 0.0309 \\ P_{1001} &= 0.0298 \\ P_{0110} &= 0.0573 \\ P_{0101} &= 0.0349 \\ P_{0011} &= 0.0853 \\ P_{0111} &= 0.1190 \\ P_{1101} &= 0.0600 \\ P_{1110} &= 0.0736 \\ P_{1011} &= 0.1143 \\ P_{1111} &= 0.2134 \\ \hline \text{Total} &= 1 \end{aligned}$$

After this, we will try to constitute the objective function with known states probabilities, so we coded the objective function and the related constraints using LINGO 8, and to export the above quantities to LINGO, we employed Ms Excel Spreadsheet. Running the coded model in LINGO revealed these solution amounts for our decision-coded variables as follows:

Table 2 Sensitivity analysis of responsiveness against n and $\bar{\lambda}$ ($\bar{\mu} = 0.037, L = 4$)

u	z	m	n	$\bar{\lambda}$	0.55	0.73	0.82	1	1.22	1.5	1.71	1.8	1.9	2
2	2	32	4	30	27	25	22	20	18	15	12	11	10	
3	2	44	5	35	33	30	27	23	21	19	14	13	11	
4	2	56	6	38	36	34	30	25	24	22	18	16	14	
4	3	76	7	44	41	39	35	31	26	23	21	19	17	
5	3	92	8	50	47	45	38	34	30	25	22	20	18	
5	4	116	9	55	52	49	43	39	35	29	26	23	21	
6	4	136	10	61	58	55	45	41	40	36	30	27	24	
6	5	164	11	65	62	60	53	46	42	41	35	32	26	
7	5	188	12	73	69	65	59	54	47	43	39	34	30	
7	6	220	13	76	72	70	63	60	55	48	44	38	35	

$$\begin{aligned} Y_{2,1} &= 1 \\ Y_{2,2} &= 1 \\ Y_{2,3} &= 1 \\ Y_{2,4} &= 1 \\ Y''_{1,1,2,1} &= 1 \\ Y''_{1,1,2,2} &= 1 \\ Y''_{1,1,2,3} &= 1 \\ Y''_{1,1,2,4} &= 1 \end{aligned}$$

5.3 Solution analysis

As shown, any request of any region for BSV will respond with BSV 2 and request for a BSV and two ASVs will respond with BSV 1, and the two ASVs numbers 1 and 2, and the other requests are all rejected. So what kind of EMS center is? The first look at the solution may be confusing, but this is the best

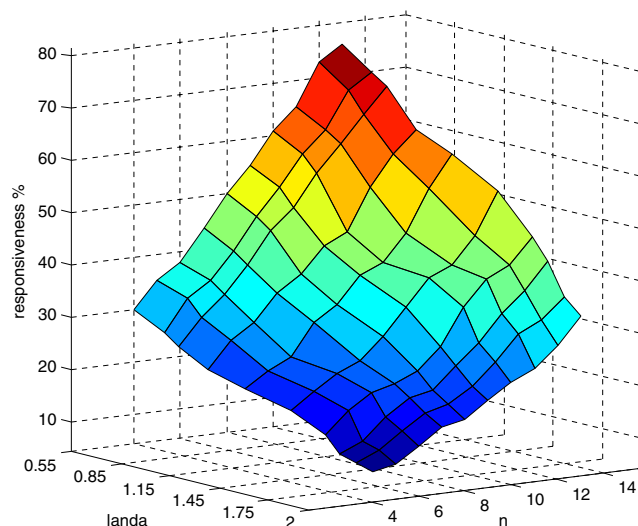


Fig. 3 Sensitivity analysis of responsiveness against n and $\bar{\lambda}$

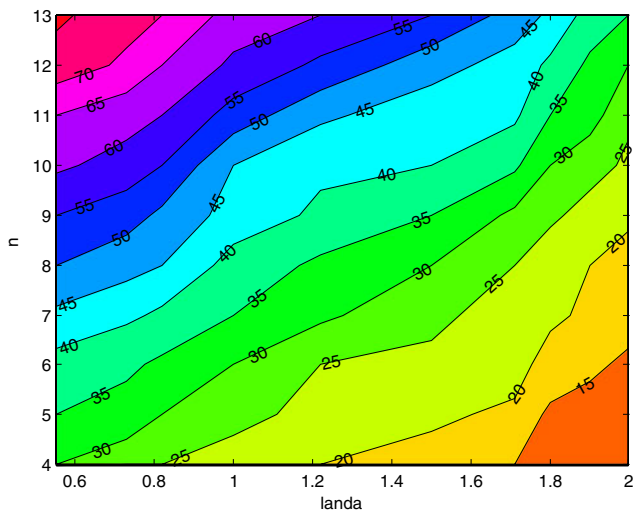


Fig. 4 Contour lines of responsiveness against $\bar{\lambda}$ and n

response which makes the coverage for the zone maximum, which means that with two BSVs and two ASVs, we cannot have unusual expectations. Looking at the model again will reveal that we had 8 kinds of requests from 4 regions, in total 32 kinds of request, and the solution is to respond to 8 kinds of requests. This is all that we could do with just four vehicles. But what is obvious is that the EMS system's responsiveness is low (25 %) due to the few number of BSV and ASV in EMS center. Certainly, system responsiveness as an important performance measure for our model would be better with more vehicles. The next section presents sensitivity analysis of responsiveness against parameters of the model.

6 Sensitivity analysis

For the purpose of sensitivity analysis, we need to present some notations as follows:

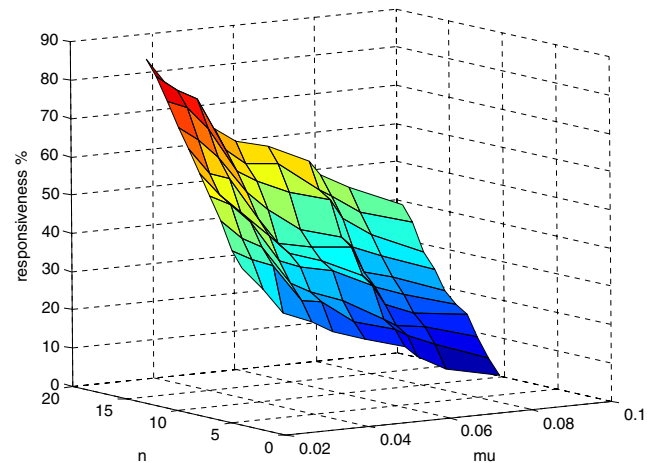


Fig. 5 Sensitivity analysis of responsiveness against n and $\bar{\mu}$

- n Total number of ambulance
- m Total number of requests
- m' Number of responded requests

Total number of ambulance is the sum of total number of BSVs and ASVs, i.e., $n = u + z$.

Total number of requests is computed from following equation:

$$m = ((u + 1) \times (z + 1) - 1) \times L$$

Where L is the number of regions. This is because for u BSVs, there may be $u + 1$ kinds of request, i.e., request for zero, one, two, three, ..., and u BSVs simultaneously. Similar statement for the z ASVs is true. For the combined request of ASVs and BSVs, all combinations are $(u + 1) \times (z + 1) - 1$, since one combination is not really a request, i.e., request for zero BSV and zero ASV. Finally, multiplying this number at the number of regions leads to the total number of requests from all regions. To compute the system's responsiveness, it is required to have the number of responded requests at a

Table 3 Sensitivity analysis of responsiveness against n and $\bar{\mu}$ ($\bar{\lambda} = 0.82$, $L = 4$)

u	z	m	$\bar{\mu}$ n	0.020	0.025	0.030	0.037	0.042	0.050	0.060	0.063	0.070	0.083
2	2	32	4	41	35	28	25	22	19	16	13	9	6
3	2	44	5	45	42	40	30	29	23	20	18	15	10
4	2	56	6	52	47	44	34	31	28	22	20	19	15
4	3	76	7	58	55	50	39	37	35	30	25	22	20
5	3	92	8	63	57	52	45	40	37	32	29	26	22
5	4	116	9	69	62	60	49	42	40	39	32	30	25
6	4	136	10	74	70	65	55	50	44	41	38	35	30
6	5	164	11	80	78	72	60	57	53	50	45	40	34
7	5	188	12	86	82	79	65	61	60	52	50	45	41
7	6	220	13	90	83	77	70	66	64	59	55	51	45

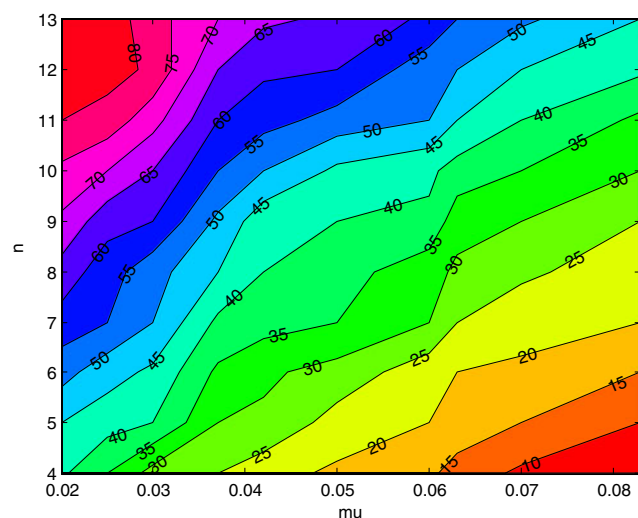


Fig. 6 Contour lines of responsiveness against n and $\bar{\mu}$

solution (m'). That is the number of variable Y in the solution with value 1. Having these two values, responsiveness is calculated from the following relation:

$$\text{responsiveness \%} = \frac{m'}{m} \times 100$$

Main parameters of the model are the demand rate of regions for vehicles (λ), service time of vehicles (μ), and the number of vehicles (n). Thus three types of sensitivity analysis were carried out to demonstrate sensitivity of system's responsiveness against two of these parameters simultaneously. Since there are many parameters of demand rate and service times, and sensitivity analysis against all of them is not insightful, so an average demand rate and service time is computed as follows and used for sensitivity analysis. For the purpose of simplicity, in computation of average demand rate, only the demand rate of each region for a BSV and an ASV is used.

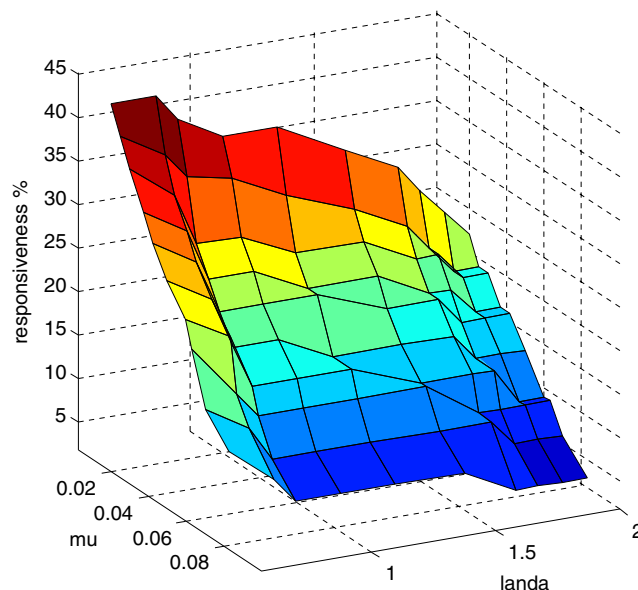


Fig. 7 Sensitivity analysis of responsiveness against $\bar{\lambda}$ and $\bar{\mu}$

$$\bar{\lambda} = \frac{\sum_{j=1}^L \lambda'_j + \sum_{j=1}^L \lambda_j}{2L}$$

$$\bar{\mu} = \frac{\mu' + \mu}{2}$$

Table 2 summarizes the results of sensitivity analysis of responsiveness against n and $\bar{\lambda}$ when $\bar{\mu} = 0.037$ and $L=4$. Figure 3 depicts the surface of responsiveness against n and $\bar{\lambda}$. As you can see, as the number of ambulance increase and average demand decrease, responsiveness increase and vice versa. Also Fig. 4 plots the contour lines of responsiveness against n and $\bar{\lambda}$. It shows an almost linear behavior in responsiveness. This plot is a useful tool for decision making as it shows tradeoffs between cost (n) and quality of the EMS center (responsiveness). For example, it determines the

Table 4 Sensitivity analysis against $\bar{\lambda}$ and $\bar{\mu}$ ($n=4, L=4, m=32$)

$\bar{\mu}$	0.020	0.025	0.030	0.037	0.042	0.050	0.060	0.063	0.070	0.083
$\bar{\lambda}$										
0.55	44	41	38	34	31	28	25	22	16	13
0.73	44	38	34	31	28	25	22	16	13	9
0.82	41	35	28	25	22	19	16	13	9	6
1	38	34	28	25	22	19	16	13	9	6
1.22	38	31	25	22	22	16	16	13	9	6
1.55	34	28	25	22	19	16	13	13	9	6
1.71	31	25	22	19	19	16	13	9	9	3
1.88	28	22	22	19	16	13	13	9	6	3
1.9	25	19	19	16	16	13	9	9	6	3
2	22	19	19	16	16	13	9	9	6	3

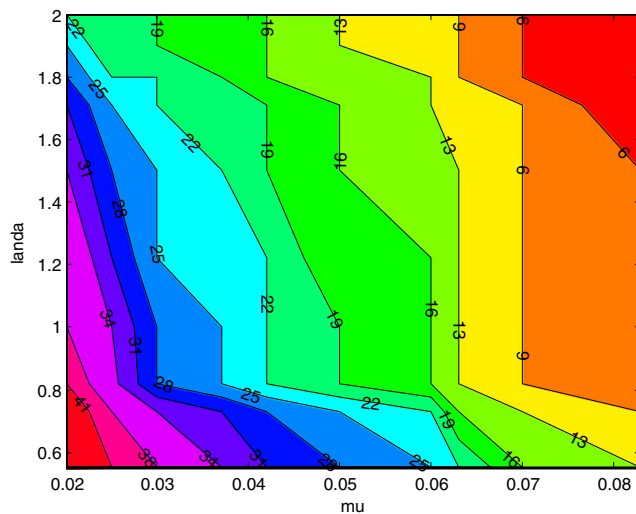


Fig. 8 Contour lines of responsiveness against $\bar{\lambda}$ and $\bar{\mu}$

number of ambulance needed to obtain a desired level of responsiveness for a specific value of average demand rate.

Table 3 summarizes the results of sensitivity analysis of responsiveness against n and $\bar{\mu}$ when $\bar{\lambda} = 0.82$ and $L=4$. Figure 5 depicts the surface of responsiveness against n and $\bar{\mu}$. One can see that as the number of ambulance increase and average service time decrease, responsiveness increase and vice versa. Also Fig. 6 plots the contour lines of responsiveness against n and $\bar{\mu}$. It shows that for different service times, adding an ambulance to the fleet has different impacts on responsiveness, i.e., for lower value of service time, the impact is greater. So, increasing the size of the fleet and improving the performance of each vehicle simultaneously is a good strategy for the EMS center.

Table 4 summarizes the results of sensitivity analysis of responsiveness against $\bar{\lambda}$ and $\bar{\mu}$ when $n=4$ and $L=4$. Figure 7 depicts the surface of responsiveness against $\bar{\lambda}$ and $\bar{\mu}$. One can see that as the average demand rate and average service time decrease, responsiveness increase and vice versa. Also Fig. 8 plots the contour lines of responsiveness against $\bar{\lambda}$ and $\bar{\mu}$. It shows that in the case of increasing the average demand rate, to keep the responsiveness from decreasing, we need to decrease the average service time. Also, it determines the amount of this decrease.

7 Conclusion and recommendation

In this paper, we discussed a new idea for introducing more than one type of emergency requests and presented a mathematical model which is made up of the features in MEXCLP and hypercube model. We applied the model for a small EMS center in TEHRAN city composed of four ambulances (two of each kind) successfully, and we concluded that the system's performance was a little low (25 %) due to the shortage of adequate

vehicles in the center (the system performance was defined as the ratio of the request calls which had been responded). Then we conducted sensitivity analysis to increase the ambulances number from each type and run the model and check the performance of the system. This analysis shows that system responsiveness varies against system parameters in a linear manner.

As a future work, the authors suggest considering more than one EMS center (decentralization plan) and then increasing the number of ambulances and finding the tradeoff, if any, between decentralization and vehicle increasing. Another research opportunity is incorporating uncertainty in the model in the form of fuzzy or stochastic parameters such as demand rates and service times. A good suggestion to run the model with more vehicles is the use of computer programming to constitute the steady-state equations automatically because of the complexity of these equations.

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